Degree: BCA, Stream: BCA
Term – I ,Third Semester Examination, September- 2026
Subject Code: BCACC301
Subject Name: Design and Analysis of Algorithms

Full Marks: 30 Set No:3 Duration: 1 Hour

Part - A
Answer any 5 questions out of 6
Each question carries 2 marks (2×5)

1. Can an algorithm have zero inputs? Justify your answer with an example. **2**

2. State the best-case and worst-case time complexities of merge sort. **2**

3. Find the dominant term and asymptotic complexity of: **2**

$$T(n) = 4n^2 + 3n + 10$$

4. Is Insertion Sort a stable sorting algorithm? Give a reason. **2**

5. State the divide-and-conquer strategy and name its three main steps. **2**

6. What is the role of a pivot in quick sort? **2**

Part - B
Answer any 2 questions out of 3
Each question carries 5 marks (5×2)

7. Consider the following algorithm: **5**

```
PROCESS(A, n)
    sum ← 0
    for i ← 1 to n
        sum ← sum + A[i]

    for j ← 1 to n
        A[j] ← A[j] * 2

    return sum
```

Using the step-count method:

a. Calculate the number of times each statement is executed.
b. Derive the total step-count function T(n).
c. Determine the asymptotic time complexity.

8. Apply Selection Sort to: **5**

A=[29, 10, 14, 37, 13, 5]

Show the array after every pass. Count the number of comparisons performed and determine whether the number of comparisons depends on the initial arrangement of the elements.

9. Consider the recurrence: **5**

$T(n)=2T(n/2) + n$, $T(1)=1$

Using the Master Theorem, determine the asymptotic time complexity of T(n). Clearly identify the values of a, b, and f(n), and state which case of the Master Theorem is applicable.

Part - C
Answer any 1 question out of 2
Each question carries 10 marks (10 × 1)

10. Solve $T(n) = 7T(n/2) + 18n^2$ using Master's theorem and recursion tree. Compare the results. **10**

11. Consider the following program segment: **10**

```
count ← 0
for i ← 1 to n
    for j ← 1 to i
        count ← count + 1
```

Answer the following:

a. Calculate the number of times the innermost statement executes.
b. Express the operation count using a summation.
c. Simplify the summation.
d. Determine the time complexity using Big-O, Big-Omega and Big-Theta.

…END…